

BDD Debugging Document

QA Automation Training Part 3 – Day 2

Simulated Failures (8 Cases)

Failure 1: Element Not Found

Scenario: Login button click fails

Steps to Reproduce:

1. Navigate to `https://the-internet.herokuapp.com/login`
2. Use incorrect selector `#loginBtn` instead of `button[type="submit"]`
3. Run the login step

Root Cause:

The selector `#loginBtn` does not exist on the page. The actual login button uses `button[type="submit"]`.

Failure Log:

```
Error: locator.click: Error: strict mode violation:
locator('#loginBtn') resolved to 0 elements
Call log:
  - waiting for locator('#loginBtn')
```

Fix: Update locator in `LoginPage.ts` to `button[type="submit"]`

Tools Used: Playwright Trace Viewer, Screenshots

Failure 2: Timeout Error

Scenario: Dashboard assertion times out

Steps to Reproduce:

1. Login with valid credentials
2. Assert `toHaveURL` with 1ms timeout
3. `await expect(page).toHaveURL('/secure', { timeout: 1 })`

Root Cause:

Timeout set too low (1ms). Page navigation takes ~500–2000ms. Playwright's default is 5000ms.

Failure Log:

```
Error: expect(received).toHaveURL(expected)
  Expected: "https://the-internet.herokuapp.com/secure"
  Received: "https://the-internet.herokuapp.com/login"
  Timeout: 1ms exceeded
```

Fix: Remove manual timeout or set to at least `10_000` ms. Rely on Playwright auto-wait.

Tools Used: Playwright Trace Viewer, Console Logs

Failure 3: Assertion Mismatch

Scenario: Login result assertion fails due to wrong expected value

Steps to Reproduce:

1. Login with `tomsmith / SuperSecretPassword!`
2. Assert login result should be `"failure"` (wrong expected value)

Root Cause:

Scenario Outline Examples table had wrong credentials (admin/admin123) which the application rejects, so result was failure instead of expected success.

Failure Log:

```
Error: expect(received).toBe(expected)
  Expected: "success"
  Received: "failure"
  at loginSteps.ts:67
```

Fix: Update Examples table with correct credentials: tomsmith / SuperSecretPassword!

Tools Used: Cucumber HTML Report, Screenshots

Failure 4: Incorrect Locator

Scenario: Form field not filled — wrong ID used

Steps to Reproduce:

1. Navigate to <https://demoqa.com/text-box>
2. Use locator #fullName instead of #userName
3. Call `fillField('name', 'John')`

Root Cause:

The actual input field ID on demoqa is #userName, not #fullName. The incorrect locator was used during initial implementation.

Failure Log:

```
Error: locator.fill: Error: strict mode violation:
  locator('#fullName') resolved to 0 elements
```

Fix: Correct locator in `FormPage.ts` to #userName

Tools Used: Chrome DevTools (Element Inspector), Playwright Codegen

Failure 5: Navigation Failure

Scenario: Page navigation fails due to wrong URL

Steps to Reproduce:

1. Call `page.goto('https://demoqa.com/textbox')` (missing hyphen)
2. Assert form is visible

Root Cause:

URL typo — correct URL is `https://demoqa.com/text-box`. The incorrect URL returns a 404 page, so all subsequent locators fail.

Failure Log:

```
Error: locator.fill: Error: strict mode violation:
locator('#userName') resolved to 0 elements
(Page is a 404 - element does not exist)
```

Fix: Correct URL in `FormPage.ts` to `https://demoqa.com/text-box`

Tools Used: Playwright Console Logs, Network Tab in DevTools

Failure 6: Missing `this.attach` — World Not Extended

Scenario: Screenshot attachment fails in After hook

Steps to Reproduce:

1. Keep `CustomWorld` as a plain class (not extending Cucumber's `World`)

2. Run a failing scenario that triggers the After hook
3. `await this.attach(screenshot, 'image/png')` is called

Root Cause:

`this.attach` is a method provided by Cucumber's base `World` class. If `CustomWorld` does not extend the `World`, `this.attach` is undefined.

Failure Log:

```
TypeError: this.attach is not a function
    at CustomWorld.<anonymous> (fixtures/hooks.ts:42:25)
```

Fix: Extend `World` from `@cucumber/cucumber` in `world.ts`:

```
import { World, IWorldOptions } from "@cucumber/cucumber";
export class CustomWorld extends World { ... }
```

Tools Used: Cucumber Report, Terminal Stack Trace

Failure 7: Step Definition Not Found

Scenario: Cucumber cannot match a Gherkin step

Steps to Reproduce:

1. Write step in feature file: When user submits the form
2. Forget to add matching step in `formSteps.ts`
3. Run tests

Root Cause:

No step definition matches the Gherkin text. Cucumber treats undefined steps as pending and marks the scenario as failed.

Failure Log:

```
? When user submits the form
  Undefined. Implement with the following snippet:
    When('user submits the form', async function () {
      // Write code here
    });
```

Fix: Add the missing step definition in `formSteps.ts`

Tools Used: Cucumber Terminal Output

Failure 8: DataTable Parsing Error

Scenario: `rowsHash()` fails on a multi-column DataTable

Steps to Reproduce:

1. Write DataTable with 3 columns (not 2):

field	value	extra
name	John	abc
2. Call `dataTable.rowsHash()` in step definition

Root Cause:

`rowsHash()` only works with exactly 2 columns (key-value pairs). A 3-column table throws an error.

Failure Log:

```
Error: rowsHash can only be called on a data table where all rows
have exactly two columns
```

Fix: Use `dataTable.hashes()` for multi-column tables, or keep the DataTable to exactly 2 columns.

Tools Used: Cucumber HTML Report, Terminal Stack Trace

Simulated Timeouts (3 Types)

Timeout 1: Page Load Timeout

Steps Used:

```
await this.page.goto("https://demoqa.com/text-box", { timeout: 500 });
```

Root Cause:

timeout: 500 (0.5 seconds) is too short for a full page load. Default is 30,000ms.

Tools: Playwright Trace Viewer → shows navigation never completed within timeout window.

Fix: Remove manual timeout or set to 30_000.

Timeout 2: Element Timeout

Steps Used:

```
await this.page.waitForSelector("#output", { timeout: 100 });
```

Root Cause:

The #output div only appears after form submission and DOM update (~300–800ms). 100ms is insufficient.

Tools: Playwright Trace Viewer → shows element wait expiry. Screenshots show blank output area.

Fix: Use `await expect(page.locator('#output')).toBeVisible()` — Playwright auto-waits up to the default timeout.

Timeout 3: Step Timeout

Steps Used:

```
setDefaultTimeout(500); // set in hooks.ts
```

Then run any step that involves a network call.

Root Cause:

`setDefaultTimeout(500)` gives Cucumber only 0.5 seconds per step. Any step with a `page.goto()` or element interaction exceeds this.

Tools: Cucumber HTML Report → step shows "Timed out" status. Console shows `Timeout of 500ms exceeded`.

Fix: Set `setDefaultTimeout(30_000)` in `hooks.ts`.

Tools Reference

Tool	Purpose
Playwright Trace Viewer	<code>npx playwright show-trace reports/traces/<name>.zip</code>
Cucumber HTML Report	<code>reports/cucumber-report.html</code> — open in browser
Screenshots	<code>reports/screenshots/</code> — auto-saved on failure
Videos	<code>reports/videos/</code> — recorded per scenario

Tool	Purpose
Console Logs	Terminal output during npm test